

OITASH GHOSH

Salt Lake, Kolkata

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EXPERIENCE

Handshake

Jul 2026 – Present

AI Evaluation Specialist (Contract)

Remote

- Engineered **40+** containerized evaluation environments with automated graders, benchmarking **AI coding agents** across concurrency debugging, security and ML infrastructure.
- Cut **agent solve rate from 80% to 40%** on a distributed systems task by redesigning it to test derived reasoning; validated each task against **3 independent implementations**, catching a **grader defect** that was failing correct submissions.

Amazon

Jan 2026 – Jun 2026

Data Engineer Intern

Bengaluru, Karnataka, India

- Built a **Python CLI** that bulk-migrated **100+** pipeline profiles, cutting deployment time by **40%**; migrated **10+** big data pipelines and resolved **Scala** build and dependency conflicts across them.
- Re-architected a **Low Inventory Audit** pipeline integrating **20+** **data sources** across 11 marketplaces, achieving **100%** output parity with the legacy system.
- Designed a **Spark/EMR** data quality framework processing **1 billion+** daily records across 5 regions, improving data reliability by **25%**.

PROJECTS

Stratakv — Distributed Key-Value Store | Go, Raft, LSM-Tree, Docker

Aug 2026

- Built a **linearizable** replicated key-value store: from-scratch **Raft** consensus over a hand-written **LSM engine** (WAL, memtable, levelled compaction, bloom filters), sustaining **1,100 ops/s** across 3 nodes with failover in **334–526 ms**.
- Proved correctness with a **self-written linearizability checker**: **3,150 operations verified across 15 leader changes** under repeated network partitions, zero anomalies; **no external dependencies**.

Order Book Matching Engine | C++20, Multithreading, CMake, Python

Aug 2026

- Implemented a price-time-priority order book sustaining **25 ns/message (39M msg/s)** at **p99 125 ns**, using flat price-level arrays, intrusive FIFOs and a lock-free feed handoff.
- Validated against an **independent reference implementation** over **12M** randomised commands with zero divergence; clean under **ASan/UBSan/TSan**.

ELD Trip Planner | React, Django, Django REST Framework, PostgreSQL

Jul 2026

- Developed a full-stack planner turning four trip inputs into an HOS-compliant route plus **auto-generated FMCSA log sheets**, from a rules engine encoding **six federal driving limits**.
- Integrated **Nominatim, OSRM and Leaflet** into a **zero-API-key** map stack, replacing rate-limited reverse-geocoding calls with an offline **30,000-city index**.

TECHNICAL SKILLS

- Languages: **Python, C++, Go, Java, C, JavaScript, SQL, Scala**
- Web & Frameworks: **ReactJS, NextJS, Redux, NodeJS, ExpressJS, Django, Tailwind CSS**
- Data & Cloud: **AWS, Spark/EMR, PostgreSQL, MongoDB, REST APIs, ETL Pipelines**
- Tools & Practices: **Git, GitHub Actions, CI/CD, Docker, Linux, Distributed Systems, LLM/Agent Evaluation, Automated Testing**

EDUCATION

Jadavpur University

2022 – 2026

B.E. Information Technology – CGPA 7.89

Kolkata, India

ACHIEVEMENTS

- Solved **1000+** coding problems across LeetCode, Codeforces, CodeChef and GFG.
- Qualified for **Round 2** of the **Adobe India Hackathon 2025**, building offline AI document intelligence under strict performance constraints.
- AIR 8120 (99.1 percentile)** in **JEE Mains**; **566th** of 100,000+ candidates (top 0.6%) in **WBJEE**.